

A Lipschitz-nonincreasing deformation of small unitary-valued maps

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Abstract

For the unit-round sphere and either the unnormalized Hilbert–Schmidt or operator-norm length metric on $U(N)$, we give an explicit strong deformation retraction of the maps with Lipschitz constant at most $1/2$ onto the constant maps. At time $t \in [0, 1]$, the Lipschitz constant is at most $(1 - t^2)/(1 + t^2)$ times its original value. The formula depends continuously on the entire map and stays in the original $U(N)$. In particular, it answers the literal question [?24](i) in Gromov’s 2017 scalar-curvature problem list, with the stronger requirement that the Lipschitz bound be preserved throughout. The proof uses a classical matrix fractional transformation and an elementary resolvent identity.

1 Statement and scope

Gromov asks in [1, p. 36], question [?24](i), whether maps $S^{n-1} \rightarrow U(N)$ with Lipschitz constant less than $1/2$ can be deformed to constant maps continuously in the input map, for every n . The preceding page specifies a standard bi-invariant metric normalized to have shortest nonconstant closed geodesics of length 2π . We state our metrics explicitly: for $\nu = \text{HS}$ or op , put

$$\|X\|_{\text{HS}} = \sqrt{\text{tr}(X^*X)}, \quad \|X\|_{\text{op}} = \sup_{\|v\|=1} \|Xv\|, \quad \ell_\nu(U) = \int \|U'(s)\|_\nu ds,$$

and let d_ν be the induced length metric on $U(N)$. The first is the usual unnormalized Riemannian metric; the second is a Finsler length metric. The Hilbert–Schmidt metric has the stated normalization; in both norms the shortest nonconstant closed one-parameter subgroups have length 2π . That normalization alone is not a definition of an arbitrary bi-invariant metric, and no such arbitrary-metric claim is intended.

Equip S^{n-1} with its unit-round geodesic distance. For $n \geq 2$, $N \geq 1$, let $\mathcal{L}_{\leq 1/2}$ denote the maps $\Phi : S^{n-1} \rightarrow U(N)$ with $\text{Lip}_{d_\nu}(\Phi) \leq 1/2$, with the uniform topology. Define $\mathcal{L}_{< 1/2}$ similarly.

Theorem 1. *Fix $p \in S^{n-1}$. Set $A = \Phi(p)$ and $W_\Phi(x) = A^*\Phi(x)$. Then*

$$H_t(\Phi)(x) = A(W_\Phi(x) + tI)(I + tW_\Phi(x))^{-1}, \quad 0 \leq t \leq 1, \quad (1)$$

is a strong deformation retraction of $\mathcal{L}_{\leq 1/2}$ onto the constant maps, and restricts to one of $\mathcal{L}_{< 1/2}$. It satisfies

$$\text{Lip}_{d_\nu}(H_t(\Phi)) \leq q_t \text{Lip}_{d_\nu}(\Phi), \quad q_t = \frac{1 - t^2}{1 + t^2}. \quad (2)$$

The terminal constant is $\Phi(p)$; every constant map is fixed at every time.

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This is a retraction onto a copy of $U(N)$, not a contraction of that copy to one prescribed point. The based subspace $\Phi(p) = I$ is contractible. No stabilization in N is needed.

2 Proof

For either norm we use unitary invariance and the ideal inequality

$$\|BXC\|_\nu \leq \|B\|_{\text{op}} \|X\|_\nu \|C\|_{\text{op}}. \quad (3)$$

Lemma 1. *If $W \in U(N)$ and $d_\nu(I, W) \leq \pi/2$, then $\text{Re } W = (W + W^*)/2 \geq 0$.*

Proof. Let $Wv = e^{i\theta}v$ with $\|v\| = 1$ and principal angle $|\theta| \leq \pi$. Every piecewise smooth unitary path U from I to W yields a path Uv on the real unit sphere underlying $\mathbb{C}^N$. Its length is at least the spherical distance $\arccos(\cos \theta) = |\theta|$, and its speed is at most $\|U'\|_{\text{op}} \leq \|U'\|_\nu$. Thus $|\theta| \leq d_\nu(I, W)$. The spectral theorem now gives the conclusion. $\square$

Write $\mathcal{S} = \{W \in U(N) : \text{Re } W \geq 0\}$. For $W \in \mathcal{S}$ set

$$F_t(W) = (W + tI)(I + tW)^{-1}.$$

The identity

$$(I + tW)^*(I + tW) = (1 + t^2)I + t(W + W^*) \geq (1 + t^2)I \quad (4)$$

proves invertibility, even at $t = 1$, and $\|(I + tW)^{-1}\|_{\text{op}} \leq (1 + t^2)^{-1/2}$. Also $(W + tI)^*(W + tI) = (I + tW)^*(I + tW)$, so $F_t(W)$ is unitary. Direct substitution gives

$$F_0(W) = W, \quad F_1(W) = I, \quad F_t(I) = I. \quad (5)$$

For $U, V \in \mathcal{S}$ the noncommutative resolvent identity is

$$F_t(U) - F_t(V) = (1 - t^2)(I + tU)^{-1}(U - V)(I + tV)^{-1}. \quad (6)$$

Indeed, multiplying on the left by $I + tU$ and on the right by $I + tV$ reduces it to

$$(U + tI)(I + tV) - (I + tU)(V + tI) = (1 - t^2)(U - V).$$

Here U commutes with its own resolvent; U and V need not commute. Equations (3)–(6) imply

$$\|F_t(U) - F_t(V)\|_\nu \leq q_t \|U - V\|_\nu. \quad (7)$$

Differentiating the same rational formula on its open domain gives

$$DF_t(W)[E] = (1 - t^2)(I + tW)^{-1}E(I + tW)^{-1}, \quad \|DF_t(W)[E]\|_\nu \leq q_t \|E\|_\nu \quad (8)$$

when $W \in \mathcal{S}$. Hence every absolutely continuous curve $W(s)$ in $\mathcal{S}$ satisfies

$$\ell_\nu(F_t \circ W) \leq q_t \ell_\nu(W). \quad (9)$$

This uses the ordinary finite-dimensional chain rule almost everywhere. For the induced length metric, the length of an absolutely continuous unitary curve is the integral of its matrix-norm speed. In particular, (9) is an intrinsic length statement, not an inference from a chordal distance bound on arbitrary pairs.

Let $L = \text{Lip}_{d_\nu}(\Phi) \leq 1/2$. Bi-invariance and the diameter π of the sphere yield

$$d_\nu(I, W_\Phi(x)) = d_\nu(\Phi(p), \Phi(x)) \leq Ld(p, x) \leq \pi/2.$$

Thus $W_\Phi(x) \in \mathcal{S}$ by Lemma 1. For x, y choose a minimizing spherical geodesic γ between them. The restriction $\Phi \circ \gamma$ is a Lipschitz, hence absolutely continuous, curve, of length at most $L\ell(\gamma)$. Its matrix-norm absolute continuity follows also from $\|U - V\|_\nu \leq d_\nu(U, V)$. Its normalized image stays in $\mathcal{S}$, so (9) and left invariance give

$$d_\nu(H_t(\Phi)(x), H_t(\Phi)(y)) \leq \ell_\nu(H_t(\Phi) \circ \gamma) \leq q_t L\ell(\gamma) = q_t Ld(x, y).$$

This proves (2), including for nonsmooth input maps.

It remains to check continuity in the input map. For eligible Φ, Ψ , let $\delta = \sup_x \|\Phi(x) - \Psi(x)\|_\nu$, $A = \Phi(p)$ and $B = \Psi(p)$. Then $\|A - B\|_\nu \leq \delta$ and

$$\|A^* \Phi(x) - B^* \Psi(x)\|_\nu \leq 2\delta.$$

Using (7) and unitarity gives

$$\sup_x \|H_t(\Phi)(x) - H_t(\Psi)(x)\|_\nu \leq (1 + 2q_t)\delta \leq 3\delta. \quad (10)$$

The rational formula is continuous on the compact set $[0, 1] \times \mathcal{S}$, by (4), so its dependence on t is uniform in W . Combined with (10), this proves joint continuity. Matrix-norm and intrinsic uniform topologies agree because the corresponding metrics induce the same topology on compact $U(N)$. Finally, (5) proves all retraction properties. Since $0 \leq q_t \leq 1$, the strict sublevel is also preserved. This proves Theorem 1.

3 Boundary cases and interpretation

The two-point sphere. If S^0 is assigned angular distance π , the same result holds. One value is fixed. For the other, the principal eigenangles satisfy $|\theta| \leq \pi/2$, and

$$F_t(e^{i\theta}) = \exp\left(2i \arctan\left(\frac{1-t}{1+t} \tan \frac{\theta}{2}\right)\right).$$

The derivative of the resulting angle is $(1 - t^2)/(1 + t^2 + 2t \cos \theta) \leq q_t$; at $t = 1$ the angle is identically zero. Thus each new absolute angle is at most q_t times the original. Distances to I are the Euclidean norm of principal angles for d_{HS} and their maximum for d_{op} , proving the asserted Lipschitz estimate. We do not use a path-length metric on disconnected S^0 .

Families and scope. For any topological parameter space Y and continuous $\Phi : Y \times S^{n-1} \rightarrow U(N)$ whose slices obey the bound, substitution into (1) gives a continuous family of deformations. The formula is also equivariant under constant changes of frames: $H_t(Q\Phi R) = QH_t(\Phi)R$ for $Q, R \in U(N)$. No claim about nontrivial sphere fibrations is needed or made. The bound $1/2$ answers the printed question; the preceding discussion of maps with $\text{Lip} < 1$ in [1] is a different bound.

Classical mechanism and priority. The fractional transformation is classical (see the Cayley chart in [2, §1.7.9]): if $C(W) = (I - W)(I + W)^{-1}$, then

$$C(F_t(W)) = \frac{1-t}{1+t}C(W).$$

Even the weaker assertion without preservation of a Lipschitz sublevel follows from the principal logarithm: for $L < 1$, the normalized image avoids -1 , and $A \exp((1-s) \log W_\Phi)$ deforms the map to A continuously. The point of Theorem 1 is its elementary, explicit quantitative estimate throughout the deformation for both stated length metrics. No novelty is claimed for the Cayley transform, logarithm chart, or local contractibility. A documented literature search accompanying this note found no earlier explicit statement of this quantitative retraction as an answer to [?24](i); this is not a guarantee of first priority.

Verification and provenance. The proof above is self-contained apart from standard finite-dimensional spectral and curve-length facts. Accompanying exact algebra checks and numerical stress tests are optional diagnostics, not a formal proof or a replacement for the analytic argument. The candidate proof, independent adversarial checks, literature search, and manuscript preparation used generative-AI assistance. The accompanying reports are AI-assisted internal reviews, not external peer review. This note is an unrefereed preprint.

References

- [1] M. Gromov, *101 Questions, Problems and Conjectures around Scalar Curvature*, incomplete and unedited version, October 1, 2017, pp. 35–36. [Author-hosted manuscript](#).
- [2] Y. A. Neretin, *Lectures on Gaussian Integral Operators and Classical Groups*, EMS Series of Lectures in Mathematics, European Mathematical Society, 2011, §1.7.9, Theorem 7.5, p. 42. [Author-hosted book](#).